

Lab 03: Modernizing to PaaS & Securing Secrets

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1. Objective

In this lab, you will **refactor** your cloud infrastructure by replacing a manually managed database server with a fully managed cloud database service. You will also learn how to store sensitive credentials securely using Azure Key Vault, and how to give your Web Server access to those credentials without ever typing a password in your code.

By the end of this lab you will have:

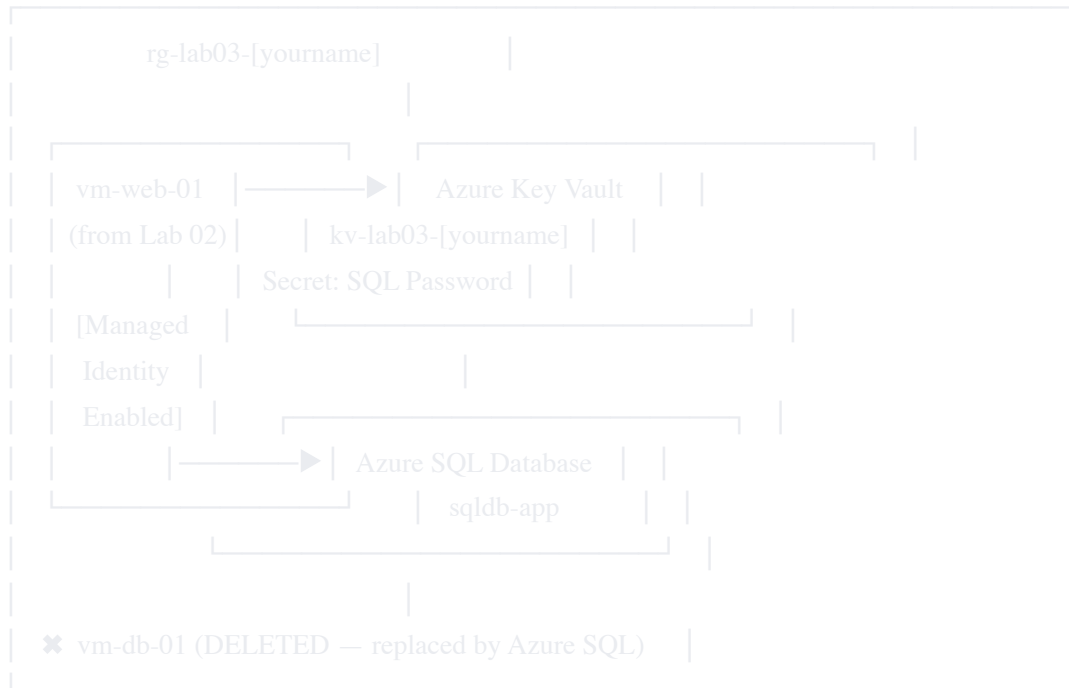
- Decommissioned the Database VM (`vm-db-01`) from Lab 02
- Deployed an **Azure SQL Database** — a fully managed Platform as a Service (PaaS) database
- Deployed an **Azure Key Vault** to store your database password securely
- Enabled a **Managed Identity** on your Web Server VM so it can retrieve secrets from Key Vault without using a hardcoded password

What does "refactor" mean in cloud infrastructure? Refactoring means improving the way something is built without changing what it does. In this case, we're swapping a self-managed database VM (where you are responsible for OS updates, backups, and availability) for a managed Azure service (where Microsoft handles all of that for you). This is a common real-world migration pattern called a "lift and modernize."

What is PaaS? PaaS stands for Platform as a Service. It means you use a cloud service without managing the underlying server, operating system, or infrastructure. Azure SQL Database is PaaS — you just use the database. Compare this to IaaS (Infrastructure as a Service), where you manage a full Virtual Machine and install the database software yourself, which is what Lab 02 used.

2. Architecture Overview

Below is a description of what your infrastructure will look like by the end of this lab:



The flow: `vm-web-01` uses its Managed Identity to retrieve the SQL password from Key Vault, then uses that password to connect to Azure SQL Database — at no point is the password stored in code or on the server.

3. Prerequisites

Complete each item below before starting the lab. Do not skip these — later steps depend on them.

- ☐ **Active Azure Subscription** — You must be able to log into <https://portal.azure.com>
- ☐ **Completed Lab 02** — `vm-web-01` must be running in `rg-lab02-[yourname]`
- ☐ **Completed Week 3 Video Modules**

How to verify Lab 02 is complete:

1. Go to <https://portal.azure.com>
2. Click **Resource groups** in the left menu
3. Open `rg-lab02-[yourname]`
4. Confirm `vm-web-01` is listed and its Status shows **Running**

If `vm-web-01` is stopped, click on it, then click **Start** and wait for it to reach the **Running** state before proceeding.

4. Naming Convention Reference

Use the table below when naming resources throughout this lab. Replacing `[yourname]` with your actual name

or initials keeps resource names unique across Azure.

Resource	Name to Use	Notes
Resource Group	<code>rg-lab03-[yourname]</code>	e.g., <code>rg-lab03-jcharles</code>
SQL Server	<code>sql-server-[yourname]</code>	Must be globally unique — all lowercase
SQL Database	<code>sqldb-app</code>	Exact name, no changes needed
Key Vault	<code>kv-lab03-[yourname]</code>	Must be globally unique
Azure Region	<code>East US</code> (preferred)	Use the closest available US region if East US is unavailable

Why do some names need to be globally unique? Resources like SQL Servers and Key Vaults get a public DNS hostname (e.g., `sql-server-jcharles.database.windows.net`). Since these names are public, no two resources across all of Azure can share the same name — similar to how two websites can't have the same domain name.

5. Step-by-Step Instructions

Phase 1: Decommission the Old Database VM

In a real-world database migration, you would migrate your data to the new platform first, verify everything works, and then shut down the old server. In this lab we are simulating that final decommission step.

Why are we deleting the VM instead of just stopping it? A stopped VM in Azure still incurs costs for its managed disk and reserved IP address. Fully deleting the VM and its associated resources is the correct practice when a resource is no longer needed.

Steps:

1. In the Azure Portal, click **Resource groups** in the left-hand navigation menu
2. Click on `rg-lab02-[yourname]` to open it
3. Locate and note the following three resources associated with `vm-db-01`:
 - The Virtual Machine itself: `vm-db-01`
 - Its OS disk (will be named something like `vm-db-01_OsDisk_...`)
 - Its Network Interface (will be named something like `vm-db-01-nic` or `vm-db-01VMNic`)
4. Click the checkbox next to each of those three resources to select them

5. Click **Delete** at the top of the resource list
6. When prompted to confirm, type `delete` in the confirmation box and click **Delete**

⚠ **Critical:** Do **NOT** delete the Resource Group or `vm-web-01`. You only want to remove the three resources tied to `vm-db-01`. If you are unsure which disk or NIC belongs to the DB VM, click on each resource and check the **Overview** tab — it will show the associated VM name.

✅ **Verify:** Refresh the resource group. `vm-db-01`, its disk, and its NIC should no longer appear. `vm-web-01` and its resources should still be present.

Phase 2: Deploy Azure SQL Database (PaaS)

You will now create a managed Azure SQL Database to replace the VM you just deleted.

What is Azure SQL Database? Azure SQL Database is a fully managed relational database service built on Microsoft SQL Server. Unlike running SQL Server on a VM, you don't manage the underlying operating system, apply patches, or configure backups — Azure does all of that automatically. You just create the database and start using it.

What is the difference between a SQL Server and a SQL Database in Azure? In Azure, a **SQL Server** is a logical management container — it holds settings like the admin login, firewall rules, and region. A **SQL Database** is the actual database that lives inside that server. You must create the server first, then create databases inside it. One SQL Server can host multiple databases.

Steps:

1. In the top search bar of the Azure Portal, type `SQL databases`
 - You will see **two similar options** appear in the search results:
 - **SQL databases** ← ✅ Click this one
 - **SQL servers** ← Do not select this

What is the difference? **SQL databases** is the Azure SQL Database service — the fully managed PaaS database this lab uses. This is your starting point.

SQL servers in the Azure Portal refers to a logical management container that holds admin credentials and firewall rules for your database. It is not a standalone product you deploy on its own. You do not need to navigate here directly — when you create a SQL Database in the next steps, the portal will prompt you to create the SQL Server container as part of the same setup wizard automatically.

2. Click the **dropdown arrow** on the **+ Create** button — do not just click the button itself. You will see two options appear:
 - **SQL database** ← ✅ Select this one
 - **SQL database (Free offer)**

Which option should I choose? Select **SQL database** — the standard option. Here's why:

The **Free offer** sounds appealing but has meaningful limitations — it is capped at 32 GB of storage, only one free database is allowed per subscription at a time, and it does not support the Basic DTU-based pricing tier you will configure in the next steps. The Basic tier is the lowest-cost option for a standard database (~\$5/month or free with Azure credits) and is what this lab is designed around.

Exception: If your Azure subscription is a free trial account and the standard **SQL database** option is blocked or shows unexpectedly high pricing, you may use **SQL database (Free offer)** as a fallback — but be aware the Compute + Storage configuration step will look different and some tier options may not appear.

Basics Tab:

What you will see: The Create SQL Database form opens on a **Basics** tab. Regardless of which Create option you selected, you will likely see a green banner at the top that reads "**Free offer applied! You get 100,000 vCore seconds, 32GB of data...**"

⚠ You must fully remove this offer before doing anything else. While the Free offer is applied, the DTU-based pricing tier option is not available inside **Configure database** — you will only see the General Purpose Serverless tier, which is not what this lab uses.

To remove it: Look for a button inside or near the banner. Depending on how you arrived at this page it will say either "**Advanced configuration**" or "**Remove offer**" — both do the same thing. Click whichever one you see.

After clicking it, confirm the banner is completely gone from the top of the page before proceeding. If the green banner is still visible with a "**Remove offer**" button, click that button now. The page will reload and the Compute + storage section will update — you are ready to proceed only once the banner has disappeared entirely.

3. **Subscription** — Leave as your default subscription
4. **Resource group** — Click **Create new**, enter `rg-lab03-[yourname]`, click **OK**
5. **Database name** — The field may be auto-filled with a generated name like `free-sql-db-1234567`. **Clear this field** and enter `sqldb-app`

⚠ Common mistake: Do not enter the server name here. The database name and the server name are two different things. The server name goes in the Server field below — this field is only for the database name. Using the correct name `sqldb-app` here is important because later phases reference it by name. If you accidentally give the database the same name as your server, you will need to delete the database and recreate it — Azure does not support renaming databases after creation.

6. **Server** — The dropdown will show "**No servers found in the selected resource group**" — this is expected since we have not created one yet. Click **Create new** directly below the Server dropdown. A **Create SQL Database Server** panel will open on the right. Fill it in as follows:
 - **Server name:** `sql-server-[yourname]` (all lowercase, no spaces)

- You will see `.database.windows.net` appended automatically — this is the public hostname for your server and is expected
- Location:** Change this to `(US) East US` if it is available in your subscription

My region options look different — is that okay? Yes. The regions available for Azure SQL Server vary depending on your subscription type. If **East US** is not listed or is greyed out, select the closest available US region (e.g., **Central US**, **West US**). The lab will work the same regardless of which US region you choose. The important thing is that you note which region you selected, as ideally all your lab resources should be in the same region to avoid latency and potential cross-region charges.

Authentication — Important: You will see a blue info banner stating "**Azure Active Directory (Azure AD) is now Microsoft Entra ID.**" This is just a name change Microsoft made — they refer to the same identity service. You do not need to do anything with this banner. Below that, you will see three **Authentication method** radio button options. The portal may have any of these pre-selected depending on your subscription type:

Option	What it does	Use for this lab?
Use Microsoft Entra-only authentication	Disables username/password login entirely	✗ No
Use both SQL and Microsoft Entra authentication	Enables both methods simultaneously	✗ No
Use SQL authentication	Enables username/password login only	✓ Yes

Select **Use SQL authentication**.

Why not "Use both"? Although "Use both" also exposes the username and password fields and would technically work, it adds Entra identity management on top of SQL credentials — unnecessary complexity for this lab. Selecting **Use SQL authentication** keeps things simple and focused on what this lab is teaching.

Once you select **Use SQL authentication**, three fields will appear:

- Server admin login:** `sqladmin`
- Password:** Create a strong password and **write it down** — you will need it in Phase 4
 - Password requirements: minimum 8 characters, must include uppercase, lowercase, a number, and a special character (e.g., `Lab03@Secure!`)
- Confirm password:** Re-enter the same password

Click **OK** to close the Create SQL Database Server panel and return to the main form.

7. Want to use SQL elastic pool? — Select No

8. **Workload environment** — Select **Development**

⚠ **This field defaults to Production — you must change it.** The Production workload environment defaults to the Hyperscale tier which costs \$320+/month. Selecting **Development** sets a lower-cost starting configuration and makes the Basic DTU tier available in the Configure database step. This field may only appear when selecting an existing server rather than creating a new one.

9. Under **Compute + storage** you will see the current tier listed as "**General Purpose - Serverless**" with a **Configure database** link below it. Click **Configure database**.

Don't see a DTU option inside Configure database? This means the free offer is still applied. Go back to the top of the Basics tab, click **Remove offer** in the green banner, wait for the page to reload, and then click **Configure database** again. The DTU-based option will only appear once the free offer has been fully removed.

10. At the top of the configuration panel, look for the purchasing model selector and choose **DTU-based** if it is not already selected

What is a DTU? DTU stands for Database Transaction Unit — a bundled measure of compute, memory, and I/O capacity. The Basic tier provides 5 DTUs, which is more than enough for a lab environment.

11. Select **Basic** from the service tier options
12. Confirm the Cost summary on the right updates to show approximately **\$4.99/month**
13. Click **Apply** — you will be returned to the Basics tab

Behavior When Free Offer Limit Reached:

13. You may see a section titled "**Behavior when free offer limit reached**" with two radio button options. If this section appears, select **Auto-pause the database until next month**

This ensures that if free tier limits are hit, the database pauses rather than continuing to run and charging you at full rates.

Backup Storage Redundancy:

14. Under **Backup storage redundancy**, select **Locally-redundant backup storage (LRS)**

Why LRS for a lab? Locally-redundant storage replicates your backups within a single Azure datacenter — the lowest-cost option and perfectly sufficient for lab use. In production you would typically select Zone-redundant or Geo-redundant storage for higher availability.

Networking Tab:

15. Click the **Networking** tab at the top of the form

You will notice the wizard now has multiple tabs: **Basics, Networking, Security, Additional settings, Tags, and Review + create**. The lab will walk you through the ones that require changes. For any tab not mentioned, you can leave all settings at their defaults.

16. **Connectivity method** — This defaults to **No access**. Change it to **Public endpoint**

What does "No access" mean? No access means the database is completely unreachable from outside Azure — not even from your own computer. We need to change this to Public endpoint so the database can be reached both by Azure resources and by you directly for management.

17. Once you select **Public endpoint**, two additional toggle options will appear below:

- **Allow Azure services and resources to access this server** — Set to **Yes**

This allows other Azure services (like your Web VM) to connect to the SQL Server over Azure's internal network backbone.

- **Add current client IP address** — Set to **Yes**

This adds a firewall rule for your current public IP address so you can connect to and manage the database from your own computer.

18. **Connection policy** — Leave as **Default**

19. **Minimum TLS version** — Leave as **TLS 1.2**

Security Tab:

20. Click the **Security** tab

21. **Enable Microsoft Defender for SQL** — Select **Not now**

Microsoft Defender for SQL provides vulnerability assessments and threat detection. It is a valuable tool in production environments but costs \$15/server/month. For this lab, we skip it to avoid unnecessary charges. You can always enable it later.

22. Leave all other settings on this tab at their defaults

Additional Settings Tab:

23. Click the **Additional settings** tab

24. **Use existing data** — Leave as **None** (we are creating a blank database)

25. **Collation** — Leave as the default `SQL_Latin1_General_CP1_CI_AS`

What is collation? Collation defines the rules for how the database sorts and compares text — for example, whether it treats uppercase and lowercase letters as the same. The default setting is appropriate for most English-language applications.

26. **Maintenance window** — Leave as **System default (5pm to 8am)**

27. Click **Review + create**

Review + Create:

28. Review the summary page and confirm:

- Tier shows **Basic** (not General Purpose or Standard)
- Estimated monthly cost is approximately **\$4.99** (or \$0.00 if covered by free credits)
- Server name matches `sql-server-[yourname]`

- Resource group shows `rg-lab03-[yourname]`

29. Click **Create** and wait for the deployment to complete (approximately 2–3 minutes)

✅ **Verify:** Once deployment completes, click **Go to resource**. The database Overview page should show the server name, database name `sqladb-app`, and status as **Online**.

Phase 3: Deploy Azure Key Vault

You will now create a Key Vault to store your SQL admin password securely.

What is Azure Key Vault? Azure Key Vault is a cloud service for securely storing and accessing secrets — things like passwords, API keys, certificates, and connection strings. Instead of storing a password in a config file or hardcoding it in an application, you store it in Key Vault and your application retrieves it at runtime. This means the password is never written down anywhere it could be accidentally exposed or committed to source control.

Steps:

1. In the top search bar, type `Key vaults` and click the result
2. Click **+ Create**

Basics Tab:

3. **Subscription** — Leave as your default subscription
4. **Resource group** — Select `rg-lab03-[yourname]` (created in Phase 2)
 - If the resource group field auto-populates with a different group, clear it and select `rg-lab03-[yourname]` from the dropdown.
5. **Key vault name** — Enter `kv-lab03-[yourname]`
6. **Region** — The portal may default to **Central US** or another region. Change this to **(US) East US**
 - Keeping all lab resources in the same region avoids cross-region data transfer costs and keeps the architecture consistent.
7. **Pricing tier** — Confirm it is set to **Standard**
 - Standard vs. Premium:** Standard covers secrets, keys, and certificates — everything needed for this lab. Premium adds Hardware Security Module (HSM) support, which is used in high-security production environments. Standard is correct here.

Recovery Options:

The Basics tab also includes a **Recovery options** section. Here is what each setting means and what to do with it:

Setting	Default value	What to do	Why
Soft-delete	Enabled (cannot be changed)	Leave as-is	Azure enforces this — deleted vaults are retained for a recovery period before permanent deletion
Days to retain deleted vaults	90	Leave as 90	Standard retention period — fine for a lab
Purge protection	Disable purge protection	Leave as Disable purge protection	Enabling this would prevent you from permanently deleting the vault during cleanup at the end of the lab

⚠ Important for cleanup: Leave **Purge protection disabled**. If you enable it, you will be unable to permanently delete the Key Vault when cleaning up at the end of the lab until the 90-day retention period expires.

Access Configuration Tab:

- Click the **Access configuration** tab
- Permission model** — You will see two options. **Azure role-based access control** is selected by default — leave it as-is 

What is Azure role-based access control (RBAC)? RBAC is Azure's central permission management system. Instead of configuring permissions directly on the Key Vault itself, you assign a standardized **role** (such as "Key Vault Secrets User") to an identity (such as your VM). That role defines exactly what actions the identity is allowed to perform. RBAC is the recommended approach because it is consistent across all Azure services and easier to audit.

What about Vault access policy? Vault access policy is an older permission model scoped only to Key Vault. It still works but is being superseded by RBAC. The portal now recommends RBAC and selects it by default.

- Resource access** — You will see three checkboxes:
 - Azure Virtual Machines for deployment** — Leave **unchecked**
 - Azure Resource Manager for template deployment** — Leave **unchecked**
 - Azure Disk Encryption for volume encryption** — Leave **unchecked**

These checkboxes grant other Azure services permission to retrieve certificates from this vault for specific infrastructure scenarios. None of them apply to this lab.

- Leave all other settings on this tab at their defaults

Networking Tab:

- Click the **Networking** tab
- Confirm the following settings match exactly — no changes are needed on this page:

- **Enable public access** — Checked 
- **Allow access from** — **All networks** selected 
- **Private endpoint** — Leave empty, do not add anything here

Why "All networks" for a lab? Allowing access from all networks means any resource — including your VM and your own computer — can reach the Key Vault over the public internet, authenticated by RBAC permissions. In a production environment you would restrict this to **Selected networks** or use a **Private endpoint** to keep all traffic on Azure's internal network. For this lab, All networks keeps setup simple without compromising security, since access is still controlled by the RBAC role assignment you will configure in Phase 5.

 **Note:** The portal displays a warning that "Traffic from all public networks can access this resource. This is not recommended for private applications or environments." This is expected and acceptable for a lab. Do not let this warning concern you.

Review + Create:

14. Click **Review + create**
15. Confirm the summary shows:
 - Resource group:
 - Key vault name:
 - Region: (or whichever region you selected)
 - Pricing tier:
16. Click **Create** and wait for the deployment to complete

 **Verify:** Once deployment completes, click **Go to resource**. You should see your Key Vault Overview page with the Vault URI displayed (e.g., .

Important — Grant yourself access to Key Vault before moving to Phase 4

With RBAC as the permission model, **no one has access to the Key Vault by default — including you as the person who created it**. If you navigate to Secrets right now you will see: *"The operation is not allowed by RBAC... You are unauthorized to view these contents."* This is expected behavior, not an error.

You must assign yourself the **Key Vault Administrator** role on this vault before you can create secrets in Phase 4. Follow these steps now:

1. In your Key Vault resource, click **Access control (IAM)** in the left-hand menu
2. Click **+ Add** → **Add role assignment**
3. In the **Role** tab, search for , select it, and click **Next**
4. In the **Members** tab, **Assign access to** — select **User, group, or service principal**
5. Click **+ Select members**, search for your own Azure account email address, select it, and click **Select**
6. Click **Review + assign**, then **Review + assign** again to confirm

Wait 1–2 minutes for the role assignment to propagate, then navigate to **Objects** → **Secrets**. The unauthorized message should be gone and you will be able to create secrets.

Why Key Vault Administrator for yourself but Key Vault Secrets User for the VM? The **Key Vault Administrator** role gives full management access — create, read, update, and delete secrets, keys, and certificates. You need this as the lab author to set up the vault contents. The **Key Vault Secrets User** role (assigned to the VM in Phase 5) only grants read access to secret values — the minimum the VM needs to retrieve the password at runtime. This distinction is a real-world least privilege pattern.

Phase 4: Store the SQL Password as a Secret

Now that Key Vault is deployed, you will store the SQL admin password inside it.

Why store it here instead of just remembering it? In a real application, your web server's code needs to connect to the database. If you hardcode the password into the application or store it in a config file on the server, it becomes a security risk — config files can be accidentally exposed, committed to source control, or read by anyone with server access. Key Vault removes the password from everywhere except one secure, audited location.

Steps:

1. In your Key Vault resource, look at the left-hand menu
2. Under the **Objects** section, click **Secrets**
3. Click **+ Generate/Import** at the top of the page — this opens the **Create a secret** form
4. Fill in the form as follows:

Field	What to enter
Upload options	Leave as Manual
Name	SqlAdminPassword
Secret value	Enter the SQL admin password you created in Phase 2
Content type (optional)	Leave blank
Set activation date	Leave unchecked
Set expiration date	Leave unchecked
Enabled	Leave as Yes
Tags	Leave as 0 tags

Secret name rules: Key Vault secret names can only contain letters, numbers, and hyphens. No spaces or special characters in the name itself — only in the secret value.

What are activation and expiration dates? These optional settings let you schedule when a secret becomes active or automatically expires. In production environments, expiring secrets on a schedule is a good security practice — it forces regular rotation of passwords. For this lab we leave these unchecked and manage the secret manually.

5. Click **Create**

✓ **Verify:** You should now see `SqlAdminPassword` listed on the Secrets page with a status of **Enabled**. Click on it, then click the current version to confirm the secret was saved. You will see metadata but not the secret value itself — Key Vault intentionally hides the value after storage.

Phase 5: Enable Managed Identity & Grant Key Vault Access

This phase has two parts. First, you will enable a **Managed Identity** on your Web VM. Then, you will grant that identity permission to read secrets from Key Vault.

What is a Managed Identity? A Managed Identity is an automatically managed identity in Azure Active Directory (Azure AD) that is assigned directly to an Azure resource — in this case, your VM. Instead of creating a username and password for your VM to authenticate with other services, Azure manages a secure credential behind the scenes. Your VM can prove "I am `vm-web-01`" to other Azure services without any password being stored anywhere.

Think of it like a building access badge that is permanently attached to a person — they don't need to remember a PIN, the badge proves who they are automatically.

System-assigned vs. User-assigned Managed Identity: A **system-assigned** identity is tied to a single resource and deleted automatically when that resource is deleted — which is what we use here. A **user-assigned** identity is a standalone resource that can be shared across multiple services, useful in more complex architectures.

Part A — Enable Managed Identity on the Web VM:

1. In the Azure Portal, navigate to **Resource groups** → `rg-lab02-[yourname]`
2. Click on `vm-web-01`
3. Find **Identity** in the left-hand menu. You can either:
 - Scroll down to the **Security** section in the left menu and click **Identity**, or
 - Type `identity` in the search box at the top of the left menu to jump to it directly ← faster
4. You will see two tabs — **System assigned** and **User assigned**. Stay on the **System assigned** tab
5. Change the **Status** toggle from **Off** to **On**

6. Click **Save** at the top of the page
7. A confirmation dialog will appear explaining that a system-assigned identity will be registered in Azure AD — click **Yes**

✓ **Verify:** After saving, the page will display an **Object (principal) ID** — a unique identifier for your VM's identity in Azure AD. Copy this value and keep it handy — it confirms the identity was created successfully.

Part B — Grant the VM Access to Key Vault Using RBAC:

Since we selected Azure RBAC as the Key Vault permission model in Phase 3, we grant access by assigning a **role** to the VM's Managed Identity directly on the Key Vault resource.

What is a role assignment? A role assignment connects three things: a **who** (the VM's Managed Identity), a **what** (a built-in role defining allowed actions), and a **where** (the Key Vault resource). The role we will assign is **Key Vault Secrets User** — which grants read-only access to secret values, exactly what the Web VM needs.

1. Navigate to your Key Vault `kv-lab03-[yourname]`
2. In the left-hand menu, click **Access control (IAM)**

What is IAM? IAM stands for Identity and Access Management. This is the RBAC control panel for any Azure resource — it shows you who has what level of access, and it's where you add new role assignments.

3. Click **+ Add** at the top, then select **Add role assignment** from the dropdown

Role Tab:

4. In the search box, type `Key Vault Secrets User`
5. Select **Key Vault Secrets User** from the results
6. Click **Next**

Why this role specifically? The Key Vault Secrets User role grants permission to **read** secret values — nothing more. It cannot create, update, or delete secrets. This follows the principle of least privilege — granting only the minimum access required for the task.

Members Tab:

7. **Assign access to** — Select **Managed identity**
8. Click **+ Select members**
9. In the panel that opens:
 - **Subscription** — Leave as your default subscription

- **Managed identity** — Select **Virtual machine** from the dropdown
- You should see **vm-web-01** appear in the list below — click on it to select it

10. Click **Select** to confirm

11. Click **Next**

If **vm-web-01** does not appear in the list, go back and confirm you completed Part A — specifically that you clicked **Save** and that an Object (principal) ID was generated on the Identity page. Wait 60 seconds for the identity to propagate and try again.

Note: You will notice a **Conditions** tab in the wizard header — it will be greyed out and the wizard will skip it automatically when assigning to a Managed Identity. This is expected, no action is needed.

Review + Assign:

12. Review the summary page and confirm the following before proceeding:

Field	Expected value
Role	Key Vault Secrets User
Scope	Should contain kv-lab03-[yourname] in the path
Members — Name	vm-web-01
Members — Type	Virtual machine

13. Click **Review + assign**, then click **Review + assign** again to confirm

 **Verify:** Click the **Role assignments** tab on the Access control (IAM) page. You should see **vm-web-01** listed with the role **Key Vault Secrets User** and Type **VirtualMachines**.

Phase 6: Validate with Azure Monitor

Now that your SQL Database is deployed and your Key Vault is configured, you will use Azure Monitor to confirm the database is live and observable.

What is Observability? Observability is the ability to understand the internal state of a system by examining its outputs — metrics, logs, and traces. In cloud infrastructure, setting up monitoring is not optional — it's how you know your resources are healthy, how you detect problems before users do, and how you prove to stakeholders that services are running. This step introduces you to that practice.

Steps:

1. In the top search bar of the Azure Portal, type **SQL databases** and click the result

2. On the SQL databases list page, look for **sqlldb-app** in the list and click on it directly

I only see the server name (e.g. **sql-server-charles) and not **sqlldb-app**:** This means you are looking at the SQL server resource, not the database inside it. Here's how to navigate to the database from the server:

1. Click on the server name to open the server Overview
2. In the left menu, scroll to the **Data management** section and click **SQL databases**
3. **sqlldb-app** will appear in the list — click on it

You can also use the breadcrumb trail at the top of any server page to navigate back, then drill into the database from there.

How do I know I'm on the database and not the server? Check the page header at the top. It should read **sqlldb-app** with a subtitle of **SQL database**. If it shows the server name with a subtitle of **SQL server**, you are one level too high — follow the steps above.

3. Once you are on the **sqlldb-app** database resource, check the **Pricing tier** in the Essentials section on the Overview page:

- If it shows **Basic** → proceed to step 4 and use **DTU percentage** as your metric
- If it shows **Free - General Purpose - Serverless** → proceed to step 4 and use **CPU percentage** as your metric instead

Tip: If you navigated through **Settings** → **Compute + storage** to verify your tier, you may have already noticed a **Database utilization** chart on the right side of that page showing DTU percentage live. This confirms your database is active and on the correct tier. You still need to complete the Metrics steps below — the Metrics page gives you a dedicated, configurable monitoring view that is the standard tool for observability in production environments.

4. In the left-hand menu, click **Monitoring** to expand it, then click **Metrics**

5. Configure the chart:

- **Metric:** Select **DTU percentage** from the dropdown
- **Aggregation:** Select **Max**

6. You will see a chart render — even if it shows near-zero activity, the line confirms the database is alive, reachable, and being monitored by Azure

What is a DTU and what is it used for? DTU stands for **Database Transaction Unit**. It is a blended measure that combines three things into a single number: **CPU** (processing power), **memory** (how much data can be held in fast-access storage), and **I/O** (how quickly data can be read from and written to disk). Instead of configuring each of these separately, Azure bundles them together into a DTU so you can reason about database capacity with a single value.

The Basic tier gives you **5 DTUs** — a fixed allocation. DTU percentage tells you how much of that allocation is currently being consumed. For example:

- **7% DTU** = the database is nearly idle, using very little of its capacity

- **80% DTU** = the database is under heavy load — a signal to consider scaling up
- **100% DTU** = the database is maxed out and queries may be slowing down or queuing

In production, you would set up **alerts** on DTU percentage so you are notified before the database becomes a bottleneck. For this lab, seeing any value on the chart — even near zero — confirms the database is live, correctly configured, and being monitored by Azure. That is the core observability habit this phase is building.

If DTU percentage is not available in the metric dropdown: Your database is on the Free - General Purpose - Serverless tier rather than the Basic DTU tier. Use **CPU percentage** as your metric instead — it serves the same observability purpose for this exercise. The concept being demonstrated is the same regardless of which metric you select.

 **Verify:** A chart appears with a visible line. No error messages are displayed on the Metrics page.

6. Troubleshooting

Issue	Likely Cause	Resolution
Cannot create SQL Database — creation fails	Server name is already taken by another Azure user	Change <code>sql-server-[yourname]</code> to something more unique, such as adding your birth year (e.g., <code>sql-server-jcharles1990</code>)
Database is on Free - General Purpose - Serverless tier after deployment	Free offer was not removed before creating the database	The DTU Basic tier cannot be applied after deployment on most subscription types. Do not attempt to change the tier from the Compute + storage page — the vCore options shown can cost \$400+/month if applied accidentally. Instead, proceed with the database as-is and use CPU percentage as your metric in Phase 6 rather than DTU percentage. The observability concept being demonstrated is identical. To avoid this in future labs, ensure the free offer banner is fully dismissed before configuring and creating the database
<code>vm-web-01</code> does not appear when selecting the Managed Identity for the role assignment	Managed Identity was not saved correctly or hasn't propagated yet	Return to <code>vm-web-01</code> → Identity and confirm the Object ID is present. Wait 60 seconds and try again
"You are unauthorized to view these contents" on	Your own user account has not been assigned an	Go to Key Vault → Access control (IAM) → Add role assignment → assign yourself Key Vault Administrator . Wait 1–2 minutes and refresh

Issue	Likely Cause	Resolution
Key Vault Secrets page	RBAC role on the Key Vault	
Metrics chart shows no data at all	Newly created database may take a few minutes to emit metrics	Wait 5 minutes and refresh the Metrics page
Database was created with the wrong name (e.g. same name as the server)	Database name field was not cleared before creation	Azure does not support renaming databases. Delete the incorrectly named database, then go to SQL databases → Create → SQL database. Select the existing server from the dropdown (do not create a new one), enter <code>sqladb-app</code> as the database name, and complete the wizard

7. Key Concepts Recap

Term	Definition
PaaS	Platform as a Service — a managed cloud service where the provider handles the underlying infrastructure
IaaS	Infrastructure as a Service — you manage the VM and everything on it
Azure SQL Database	A fully managed relational database service based on SQL Server
Azure Key Vault	A secure store for secrets, keys, and certificates
Secret	A sensitive value (like a password or API key) stored securely in Key Vault
Managed Identity	An Azure AD identity automatically assigned to an Azure resource — no password required
RBAC Role Assignment	Granting an identity permission to perform actions on an Azure resource by assigning a standardized role
Key Vault Secrets User	A built-in Azure role that grants read-only access to Key Vault secret values
Least Privilege	The security principle of granting only the minimum permissions needed to perform a task
Observability	The practice of monitoring metrics, logs, and traces to understand system health
DTU	Database Transaction Unit — a measure of compute, memory, and I/O capacity for Azure SQL

Term	Definition
LRS	Locally-redundant storage — data is replicated within a single datacenter

8. Lab Completion Checklist

- ☐ `vm-db-01`, its OS disk, and its NIC have been deleted from `rg-lab02-[yourname]`
- ☐ `vm-web-01` is still running in `rg-lab02-[yourname]`
- ☐ Azure SQL Database `sqldb-app` deployed on the **Basic** tier in `rg-lab03-[yourname]`
- ☐ Azure Key Vault `kv-lab03-[yourname]` deployed in `rg-lab03-[yourname]`
- ☐ Secret `SqlAdminPassword` created and **Enabled** in Key Vault
- ☐ System-assigned Managed Identity **enabled** on `vm-web-01` with an Object ID present
- ☐ Key Vault access policy created for `vm-web-01` with **Get** and **List** secret permissions
- ☐ Azure Monitor Metrics chart confirmed showing DTU percentage for `sqldb-app`

9. Clean Up Resources

When you have completed the lab and are done exploring, delete all resources to avoid ongoing charges.

Delete the Lab 03 resource group:

- Go to **Resource groups** → `rg-lab03-[yourname]`
- Click **Delete resource group**
- Type the resource group name to confirm, then click **Delete**

Deleting a resource group removes everything inside it — the SQL Server, SQL Database, and Key Vault — in a single action.

Optionally delete the Lab 02 resource group:

If you have finished with `vm-web-01` and all Lab 02 content, repeat the above steps for `rg-lab02-[yourname]`.

⚠ Do not delete Lab 02 resources if you plan to continue to Lab 04. Check the Lab 04 prerequisites before cleaning up.